

Make Your Own FM Transmitter For Broadcasting

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This is one of the simplest FM transmitters that can be built. It does not require any tuning, and transmit frequency is crystal locked to 90MHz. It eliminates frequency drift common to LC-based FM transmitters. This design is best suited for broadcasting voice or music up to a distance of fifty metres depending on the antenna used.

Circuit and working

The simple FM transmitter is built around low-power audio amplifier using LM386 (IC1), transistor PN2222A (T1), 30MHz crystal (XTAL1), varactor diode 1SV149 (D1) and a few other components. Inductor L1 is a three-turn coil made with 20SWG wire that has 8mm diameter with half-turn taping.

Tank circuit at the collector of T1 comprising inductor L1 and capacitor C5 is tuned to three times the crystal frequency, or 90MHz. Third harmonic of 90MHz is boosted and coupled to

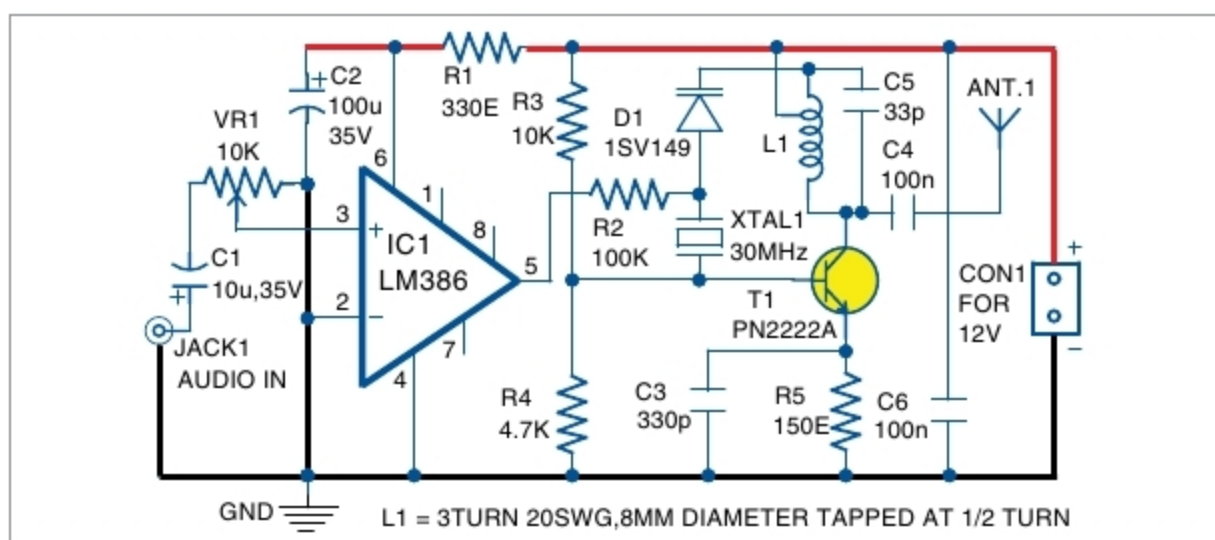


Fig. 1: Circuit diagram of the simple FM transmitter

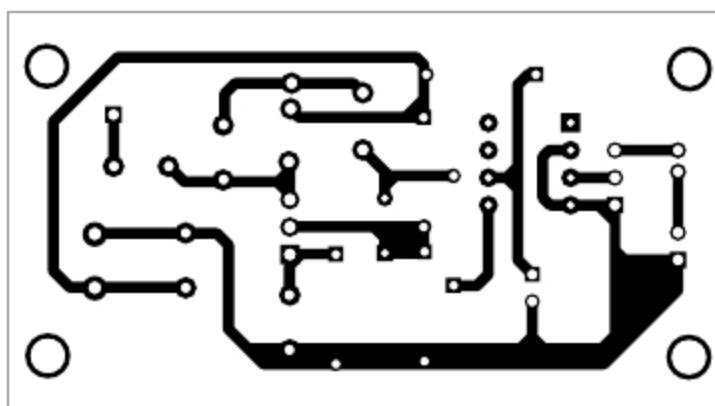


Fig. 2: Actual-size PCB layout for the simple FM transmitter

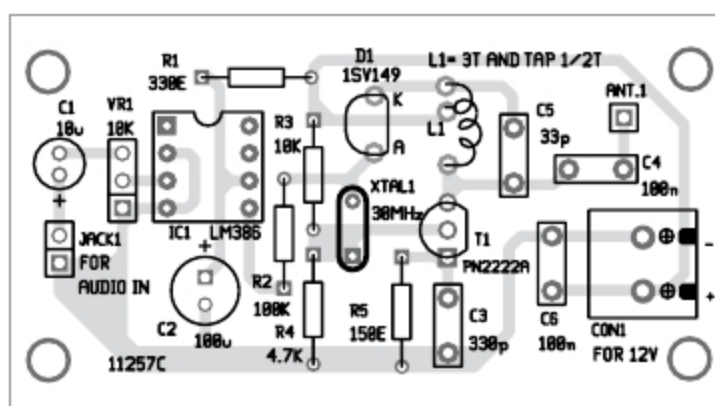


Fig. 3: Components layout for the PCB

a wire antenna via capacitor C4 for transmission that can be picked up by an FM receiver thirty to fifty metres away.

The audio amplifier formed around IC1 has a gain of twenty. Its frequency modulates the oscillator by varying the capacitance of varactor diode D1, which is connected in series with XTAL1 via resistor R2. Potmeter VR1 sets frequency deviation and should

be adjusted for best sounding audio and low distortion. The transmitter accepts audio input from such external devices as a computer, music player or cellphone. Use an 83cm-long piece of wire for the antenna.

Construction and testing

An actual-size PCB layout for the simple FM transmitter is shown in Fig. 2 and its components layout in Fig. 3. After assembling the circuit on the PCB or veroboard, enclose it in a suitable plastic box. Use a shielded cable for input audio connection for hum-free signals.

Audio quality of the transmitter is decent, and fidelity is good enough for music. Connect 12V DC regulated power supply across connector CON1.

Connect the audio input at JACK1 from a computer, music player or cellphone. You can listen to the audio on an FM radio by tuning to 90MHz frequency band. **EFY**



Joy Mukherji is an electronics hobbyist who likes designing radio frequency circuits

PARTS LIST

Semiconductors:

- IC1 - LM386 low-power audio amplifier
- T1 - PN2222A npn transistor
- D1 - 1SV149 varactor diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 330-ohm
- R2 - 100-kilo-ohm
- R3 - 10-kilo-ohm
- R4 - 4.7-kilo-ohm
- R5 - 150-ohm
- VR1 - 100-kilo-ohm potmeter

Capacitors:

- C1 - 10 μ F, 35V electrolytic
- C2 - 100 μ F, 35V electrolytic
- C3 - 330pF ceramic disk
- C4, C6 - 100nF ceramic disk
- C5 - 33pF ceramic disk

Miscellaneous:

- CON1 - 2-pin terminal connector
- JACK1 - Mono jack
- XTAL1 - 30MHz crystal
- ANT.1 - 83cm wire piece
- L1 - Refer text
- 12V DC regulated power supply or 12V battery
- PCB

(All these are available from kitsnspares.com)